KALINGA INSTITUTE OF INDUSTRIAL TECHNOLOGY
Deemed to be University U/S 3 of the UGC Act, 1956

DEPARTMENT OF TRAINING & PLACEMENT
KALINGA INSTITUTE OF INDUSTRIAL TECHNOLOGY (KIIT)
DEEMED TO BE UNIVERSITY, BHUBANESWAR (ODISHA)

No. KIIT-DU/T&P/26/565 **Date : 13th July'2026**

Kind Attenⁿ : 2027 Graduating B.Tech(CSE/CS&SE/CS&CE/IT/ETC/E&CSE/EE/EEE Mechanical/Mechatronics) Students

This is to inform all the above students that **Zaptics Research** will be conducting an Internship Cum Performance Based PPO Recruitment Drive very shortly.

Eligible interested students are directed to **REGISTER** in both the links given below on or before **14th July'2026 by 4.00 PM** to participate for the same.

Zaptics Research - Hiring 2027 – Fill out form

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https://tinyurl.com/Reg-ZapticsResearch-27

Profile Offered	:	Mechanical Engineer for B.Tech (Mechanical/ Mechatronics) Students
	:	Embedded Hardware for B.Tech (EE /ETC/E&CSE/EEE) Students
	:	Embedded Software for B.Tech(CSE/CS&SE/CS&CE/IT/ ETC/E&CSE/EE/EEE) Students
Eligibility Criteria	:	6.00 or above CGPA in B.Tech
	:	No Backlogs
Process	:	To be notified
Internship Duration	:	4 to 6 Months
Joining	:	Will be informed
Joining Location	:	On-site, Delhi / NCR
Stipend	:	Rs. 20,000 Per Month
CTC	:	Rs. 6.00 LPA

Prof. (Dr.) Prachet Bhuyan
Professor, School of Computer Engineering,
Dean (T&P), & Chairperson (TMS), KIIT-DU

NB : 1.The company will be shortlisting from the register students list
2. Find below the job descriptions for your reference

Job Description for Mechanical Engineer

Role Overview :

We are looking for a Mechanical Design Engineer with strong hands-on experience in plastics, sheet metal, and thermal engineering for automotive or electronic products. This role will own the mechanical and thermal architecture of automotive clusters,

electronic control units, and battery systems — from concept and simulation to tooling and production.

Key Responsibilities

• Design and develop plastic and sheet-metal parts for automotive clusters, ECUs, and battery systems.
• Own full mechanical and thermal architecture: packaging, mounting, sealing, cooling strategy, vibration robustness, and serviceability.
• Create detailed 3D CAD models and 2D manufacturing drawings.
• Perform thermal simulations (CFD / conduction) to validate component temperatures, airflow, and heat dissipation.
• Define thermal concepts such as heat-sinks, spreaders, vents, fans, and interface materials.
• Perform tolerance stack-ups, GD&T, and DFMEA inputs.
• Work closely with electronics teams on PCB layout constraints, connector placement, EMI considerations, and thermal paths.
• Support prototype builds, supplier discussions, and tooling kick-off.
• Review mold-flow, tooling designs, and DFM feedback.
• Support validation activities: thermal cycling, vibration, ingress protection, and vehicle integration.
• Drive mechanical changes through development and production phases.
• Maintain mechanical and thermal documentation and release packages.

Required Skills & Experience

• Strong experience in plastic injection molded parts and sheet-metal enclosures.
• Hands-on experience with thermal simulations for electronics or battery systems.
• Practical knowledge of ribs, bosses, snaps, sealing features, fastening, inserts, and assembly design.
• Understanding of heat transfer, airflow management, and electronics cooling.
• Experience creating production-grade CAD and drawings.
• Knowledge of GD&T, tolerance stack-ups, and material selection.
• Experience working with suppliers for prototypes and tooling.

Good to Have

• Experience in automotive clusters, ECUs, power electronics, or battery packs.
• Experience with IP-rated enclosures and harsh environment design.
• Exposure to automotive development processes (APQP, PPAP, DV/PV).
• Hands-on tools: SolidWorks / Creo / NX / Catia + Ansys / Icepak / FloTHERM / SimScale.
• Basic structural FEA and vibration exposure.

What You'll Get

• Ownership of real automotive and battery products end-to-end.
• Opportunity to build indigenous EV and energy systems.
• Strong technical depth in electronics packaging and thermal design.
• Fast growth into Lead Mechanical / System Architect roles.
• Startup environment with real customer exposure and product responsibility.

Job Description Embedded Hardware

Role Overview

We are looking for an Embedded Hardware Engineer who is passionate about designing robust electronic systems and taking products from concept to working prototypes and production. You will work closely with firmware, mechanical, and system teams to develop next-generation automotive and energy products.

Key Responsibilities

• Design and develop embedded hardware for automotive / EV / energy products (schematic, PCB, BoM).
• Work on microcontroller/SoC-based systems (NXP, TI, STM, ADI, etc.).
• Own board bring-up, debugging, and validation (power, boot, interfaces, signals).
• Design and test power supplies, protection circuits, sensing and communication interfaces (CAN, RS485, SPI, I2C, UART, Ethernet).
• Support prototype builds, EMS coordination, and manufacturing handover.
• Perform hardware testing, fault analysis, and design iterations.
• Collaborate with firmware teams on pin-muxing, schematics, and performance optimization.
• Participate in system architecture discussions and product definition.
• Contribute to documentation, test reports, and design justifications.

Required Skills & Experience

• Bachelor's/Master's in Electronics, Electrical, or related field.
• Strong fundamentals in analog and digital electronics.
• Hands-on experience with schematic design and multi-layer PCB design.
• Experience in board bring-up and hardware debugging using DSO, logic analyzer,
• Knowledge of microcontrollers, PMICs, ADCs, isolation, and communication protocols.
• Understanding of EMI/EMC, grounding, and automotive-grade design practices.
• Ability to read datasheets, reference designs, and apply them practically.

Good to Have

• Experience in automotive, EV, BMS, motor control, or power electronics.
• Exposure to safety-critical or high-reliability designs.
• Experience working with EMS vendors and production support.
• Familiarity with Altium / KiCad / OrCAD.
• Basic firmware understanding for bring-up support.

What You'll Get

• Opportunity to build core products, not just support projects.
• End-to-end ownership of real hardware that goes into vehicles and energy systems.
• High learning curve across automotive, power, and embedded systems.
• Direct exposure to customers, manufacturing, and system architecture.
• Startup environment with high impact, speed, and responsibility.

Job Description Embedded Software

Role Overview

We are seeking an Embedded Systems Engineer with strong experience in EV subsystem development—primarily involving instrument clusters, motor controllers, chargers, ECUs, and other such components. The role focuses on understanding

system-level requirements, developing ground-up firmware for such systems, and customisations for different clients.

The ideal candidate will work across embedded firmware, CAN communication, diagnostics, testing, and release documentation.

Key Responsibilities

1. Embedded Software & Firmware Development

• Design, develop, and optimize embedded C/C++ firmware for ESP32, STM32, and NXP microcontroller platforms (mandatory requirement).

• Develop bootloader logic, safety routines, task scheduling, and peripheral initialization for EV components.

• Implement and tune control algorithms, sensor interfaces, diagnostics, error handling, and safety checks.

• Develop firmware modules for communication interfaces: CAN, LIN, UART, SPI, I2C, ADC, and PWM-based motor-control signals.

• Work closely with hardware teams to validate firmware bring-up on new PCB prototypes.

2. Software Architecture, RTOS & Application Layer Development

• Develop system-level firmware architecture including driver layer, middleware, and application layer.

• Build real-time applications using FreeRTOS including task design, memory management, and inter-task communication.

• Implement OTA (over-the-air) / wired firmware update flows where applicable.

• Develop safety-related software mechanisms such as watchdogs, input validation, and fault recovery logic.

3. Model-Based Development (Optional but Preferred)

• Contribute to developing or validating control algorithms using MATLAB/Simulink (MIL/SIL).

• Support automation of test cases and model-to-code workflows if adopted.

4. CAN & DBC File Expertise

• Create and modify DBC files based on subsystem specifications.

• Ensure smooth communication between all EV components through correct signal mapping and diagnostics integration.

• Validate CAN messages using tools such as Vector, CANoe, CANalyzer, or equivalent.

5. Testing & Validation

• Perform HIL testing, subsystem bench testing, and full-vehicle integration validation.

• Conduct track/field tests to verify system behaviour under real-world conditions.

• Capture, analyse, and interpret CAN logs; identify root causes and validation gaps.

• Collaborate with suppliers to resolve failures and validate fixes.

6. Documentation & Release Management

• Prepare Integration Documents, CAN communication specifications, and test reports.

• Maintain firmware and configuration version control using Git/SVN.

• Deliver integration documentation aligned with project milestones.

7. Cross-Functional Coordination

• Work closely with OEM teams, component suppliers, dealerships, and validation teams.

• Drive clarity in requirements and resolve interface-level issues proactively.
• Support field issues and assist in root cause analysis for integration problems.

8. Python Scripting

• Develop Python tools for automation, CAN log decoding, test script development, and data analysis.

Required Skills & Qualifications

• Strong understanding of instrument clusters, motor controllers, chargers, and EV electrical architecture.
• Proficient in embedded C, STM32/ESP32 MCUs, and communication protocols (CAN, LIN, UART).
• Experience with CAN tools (PCAN, etc.).
• Experience with RTOS (FreeRTOS preferred).
• Hands-on exposure to HIL testing, bench testing, and vehicle validation.
• Strong documentation skills (Confluence, Word, Git/SVN).
• Python scripting experience
• Experience in the EV domain strongly preferred.

About the company

Zaptics Research (www.zaptics.io) is an innovative research and product-development company focused on designing and engineering advanced electronic control and energy storage solutions for today's automotive and energy markets.

Founded by Ather Alumni, Zaptics specializes in end-to-end development of high-reliability systems such as battery management systems (BMS), vehicle control units (VCUs), EV charge controllers (EVCC), and automotive clusters, combining deep hardware and embedded software expertise to deliver robust, production-ready solutions for electric vehicles and stationary energy storage applications.